

Performance Analysis of TCP over 802.11n with MPDU Aggregation

Recap – TCP Basics

- TCP ensures reliable transmission with:
 - **Congestion Control** (Slow Start, Congestion Avoidance, Fast Retransmit/Recovery).
 - **Flow Control.**
- Variants: Tahoe, Reno, NewReno, Cubic.
- Challenge: How does TCP perform over **wireless networks**?

IEEE 802.11n

- A **Wi-Fi standard** (2009).
- Improvements over 802.11a/b/g:
 - **MIMO** (Multiple Input, Multiple Output).
 - **Channel bonding** (20/40 MHz).
 - **Frame aggregation** (A-MPDU, A-MSDU).
 - Higher PHY rates: **up to 600 Mbps**.

MSDU



MAC Service Data Unit



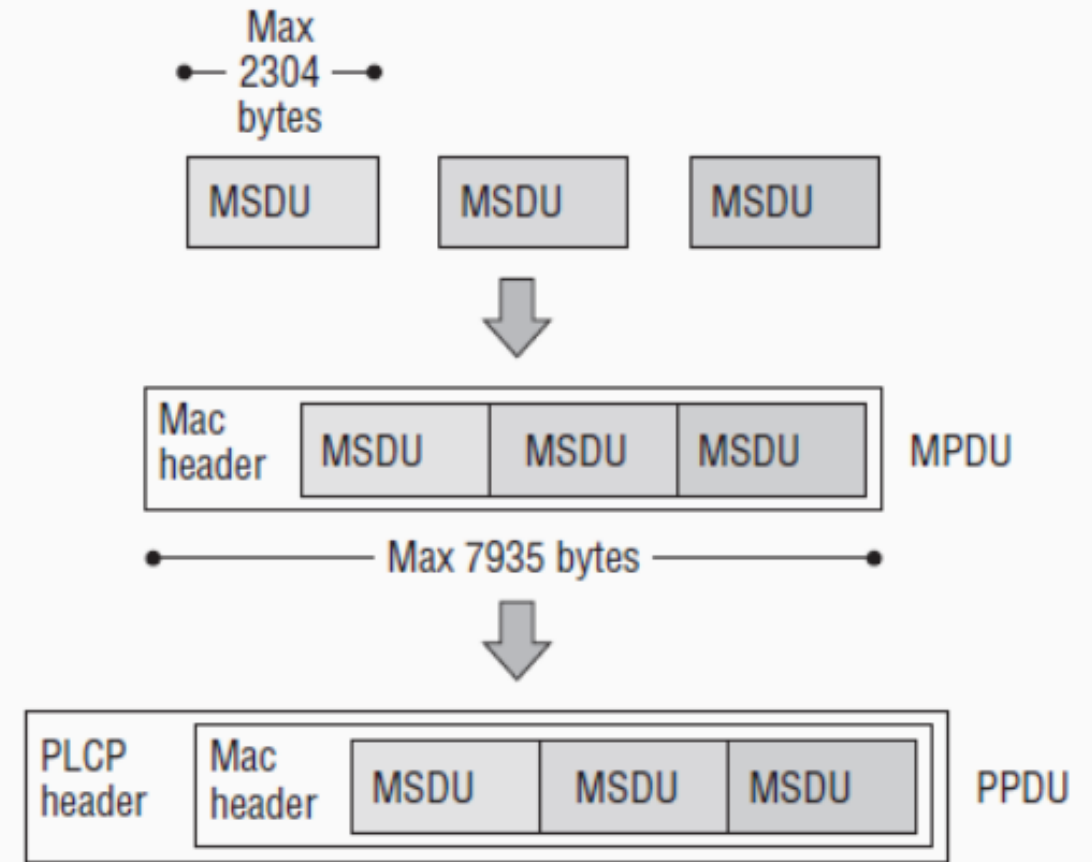
Payload from higher layers (e.g., IP packet, TCP segment)



Given to the MAC layer before adding MAC header/trailer

A-MSDU

- Aggregated MAC Service Data Unit
- Combines multiple MSDUs (higher-layer data units) into a single large payload
- Packed together within one MPDU frame for transmission
- Advantages:
 - Reduces per-packet MAC header overhead
 - Improves Wi-Fi network throughput efficiency
- Each MSDU inside the A-MSDU includes its own subframe header for separation



MPDU



MAC Protocol Data Unit



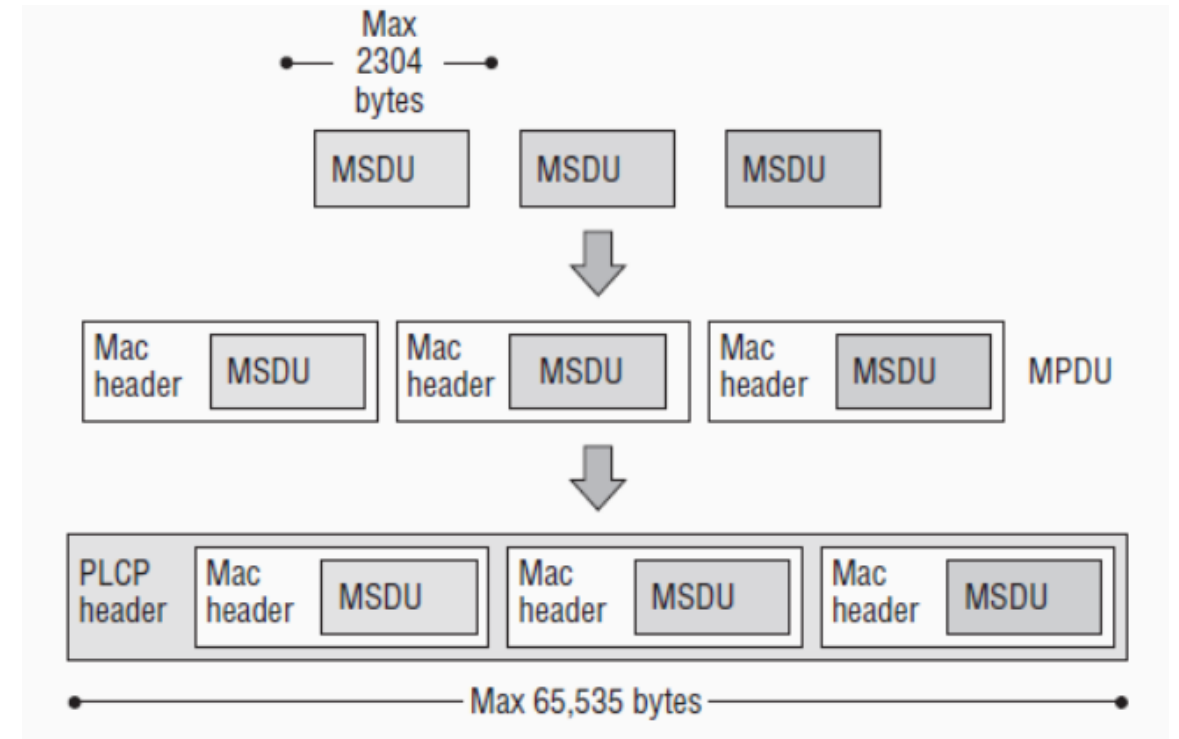
MSDU combined with MAC header and trailer



Actual frame transmitted by the MAC layer

A-MPDU

- Aggregated- MPDU
- 802.11n can **aggregate multiple MPDUs** into a single transmission.
- Benefit: Reduces overhead (headers, ACKs).
- Effect: **Higher throughput efficiency**, especially for TCP.



How it Works with TCP

- **TCP Data Transmission:** TCP transmits data in Segments.
- **Encapsulation (MSDU to MPDU):** Each TCP segment is first encapsulated as an MSDU at the MAC sublayer. When the MAC header and trailer are added, the MSDU becomes an MPDU.
- **Aggregation (A-MPDU):** In Wi-Fi, multiple MPDUs can be aggregated to form an A-MPDU, which is then sent as one large frame.
- **Transmission:** The A-MPDU is then transmitted over the Wi-Fi interface.

